

Physics Lab Manual

Course: University Physics (1)

Teacher: Wang Min

File id: mock-wjid-103 (mock=103)

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Lab 1 Measurement and error analysis

Goals: use a vernier caliper and a micrometer screw gauge; report a measurement with its absolute and relative uncertainty.

Apparatus: steel cylinder, vernier caliper, micrometer, balance.

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Lab 2 Uniformly accelerated motion

Goals: measure the acceleration of a cart on an inclined track and compare it with the value predicted from the incline angle.

Apparatus: air track, photogates, digital timer.

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Lab 3 Conservation of momentum

Goals: verify momentum conservation in a two-body collision and classify the collision as elastic or inelastic from the energy loss.

Apparatus: air track, two carts, photogates, electronic balance.

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Lab 4 Simple harmonic motion

Goals: measure the period of a spring pendulum as a function of the suspended mass and extract the spring constant from the slope.

Report: every lab report needs data tables, error bars and a short discussion of the dominant source of uncertainty.